Project Overview

The final project I worked on for my GTSF was analyzing the volatility and correlation dynamics of three assets and the correlation dynamics of two sectors, with one being an individual stock that piqued my interest because it laid off around 40% of its workforce, and two ETFs (XLE and XLK), as it's highly relevant industries in today's media. I wanted to understand whether these two dominant sectors co-move predictably enough to support pairs trading or sector-hedging strategies, especially during periods of macroeconomic stress and major global news. Both datasets were pulled from yfinance, which is just Yahoo's library for getting OHLCV data for any stock. My goal was to answer two questions. First, can we predict how volatile these assets will be tomorrow based on their recent behavior? Second, does the relationship between the Energy and Tech sectors stay consistent, or does it break down when markets get stressed?

Methodology

I started with exploratory data analysis for each asset, examining rolling 30-day price averages and volatility, return distributions, and the rolling correlation between XLE and XLK. The goal was to spot patterns before modeling to determine whether volatile days cluster together, which is exactly what the GARCH model family does, and whether the relationship between Energy and Tech remains stable over time. This step matters because the patterns you find here tell you which models are worth running. If volatility doesn't cluster, there's no point fitting a volatility model. If correlation never changes, there's no point in modeling it as time-varying.

From there, I assess different GARCH models (ARCH, GARCH, and EGARCH) to determine which one best fits the underlying stock, each adding a different component, and see how volatility carries over day by day. For the correlation between XLE and XLK, I used a DCC-GARCH model, which extends volatility models to capture how the relationship between two assets shifts over time rather than assuming it's fixed, and to assess whether sector co-movement breaks down during periods of stress. Finally, I checked the models against a naive model that just predicts "tomorrow's volatility = today's volatility," because if a complex model can't beat a simple model that makes that guess, it's not actually adding value.

Key Findings

The most interesting result came from the DCC-GARCH model for XLE and XLK. The correlation between Energy and Tech jumped from a baseline of 0.20 across the rest of

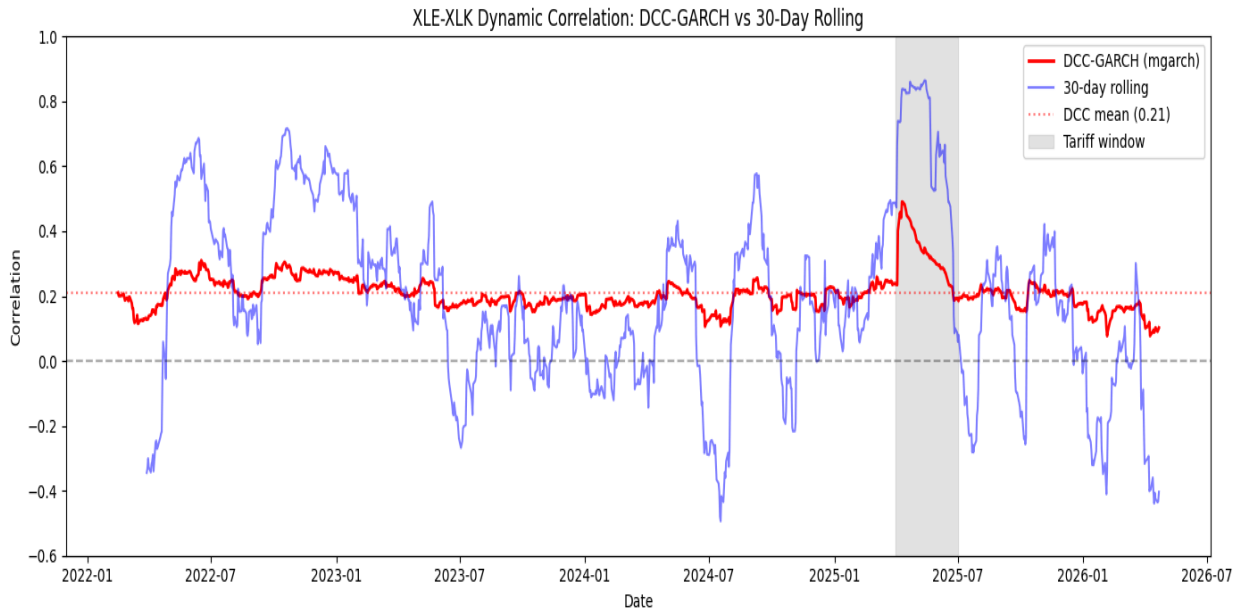
the sample to 0.33 during the 2025 U.S. tariff window, an increase of about 60%, peaking at 0.49 on April 10, 2025. This matters because the model also showed that correlation is highly persistent (the model's persistence parameter was 0.95), meaning these regime shifts last for weeks. When macro stress hits, Energy and Tech move together much more than usual, and they stay that way for a while. The real-world consequence is that any pairs trade or sector hedge built on calm-period correlation would have been badly under-hedged during exactly the period when you'd most want protection, which is the entire point of hedging in the first place.

The second finding confirmed what the EDA hinted at which was that both XLE and XLK show clear volatility clustering. Their squared returns had significant autocorrelation at lags 1, 3, and 4, which is a fancy way of saying yesterday's volatility actually does predict today's. EGARCH(1,1) beat regular GARCH on fit metrics for both ETFs, which makes sense because EGARCH accounts for the fact that bad news spikes volatility more than good news of the same size

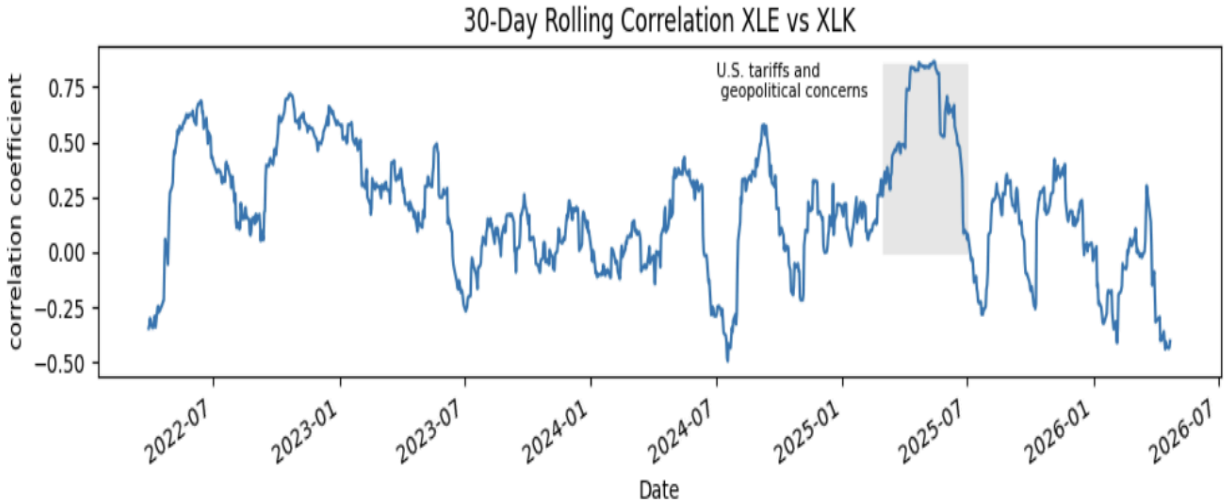
The third finding was the most surprising; Block (XYZ) behaved nothing like the sector ETFs. The EGARCH model's forecast was 31.6% worse than a naive baseline that simply assumes tomorrow's volatility equals today’s, and the model's shock-response parameter was statistically zero. The fancy model essentially provided zero value. The takeaway is that individual stocks like Block don't have the same clean, predictable volatility structure that diversified sector ETFs do. They're driven more by company-specific news (like the 40% layoff announcement) than by smooth statistical patterns.

The fourth finding was that the price spread between XLE and XLK trended steadily from −20 to −100 over the sample rather than oscillating around a stable average. This kills the most basic version of a pairs trade since there wasn't a stable mean to revert to.

Visuals



DCC-GARCH dynamic correlation (red) vs 30-day rolling correlation (blue) for XLE-XLK. The shaded region marks the 2025 U.S. tariff window where DCC correlation spiked from a baseline of 0.21 to a peak of 0.49 (~60% above baseline)



30-day rolling correlation between XLE and XLK. The relationship swings dramatically between −0.5 and +0.85 over the sample, with a clear spike during the 2025 U.S. tariff window (shaded). This instability is the primary reason DCC-GARCH was needed rather than assuming a constant correlation.

Reflection

After experimenting with many different models and seeing how volatility is a method of comparing different stocks with one another, one approach I would take moving forward is to use different types of time-series modeling, specifically ML/deep learning models, so it can try to make a more accurate prediction in forecasting. I was also surprised that a fancy GARCH model with all its statistical power wasn't able to make a more accurate prediction than a simple guessing model, showing that complexity needs to be justified with data. Lastly, I would apply this exact framework to other cross-sectors and see if there's a pattern to be uncovered, or whether pair trading could be a possibility between industries.